ANSWER SHEET 2023 SPRING

1) 20-kVA, 24,000/277-V DT, Δ-Y

LVS ⇒ $V_{line,OC} = 480V$, $I_{line,OC} = 4.05A$, $P_{3\phi,OC} = 948W$, $I_{line,SC} = 1.80A$, $P_{3\phi,SC} = 915W$, $V_{line,SC} = 1400V$, HVS

a) per unit equivalent circuit

b) VR at rated load and 0.9 PF lagging

c) η% ?

a) $|Y_{EX}| = \frac{I_{\phi,OC}}{V_{\phi,OC}} = \frac{4.05A}{277V} = 0.0146\ S$

The angle $\theta = -\cos^{-1}\left(\frac{P_{\phi,OC}}{V_{\phi,OC} I_{\phi,OC}}\right) = -\cos^{-1}\left(\frac{948/3}{(277)(4.05)}\right)$

$\Rightarrow \theta = -73.64°$

LVS ⇒ $Y_{EX} = G_C - jB_M = 0.0146\angle -73.64° = 0.004112 - j0.014$

$\Rightarrow R_C = 243.2\ \Omega$, $X_M = \frac{1}{B_M} = 71.423\ \Omega$, $V_{\phi,base} = \frac{277}{\sqrt{3}}$

$Z_{base} = \frac{(V_{\phi,base,S})^2}{S_{\phi,base}/3} = \frac{(277/\sqrt{3})^2}{20kVA/3} = 3.836\ \Omega$

$R_C = \frac{243.2}{3.836} = 63.4\ pu$. $X_M = \frac{71.423}{3.836} = 18.6\ pu$

HVS ⇒ $V_{\phi,SC} = V_{L,SC} = 1400V$, $I_{\phi,SC} = I_{L,SC}/\sqrt{3} = 1.04A$, $P_{\phi,SC} = 915/3 = 305W$

$Z_{EQ} = \frac{V_{\phi,SC}}{I_{\phi,SC}} = \frac{1400V}{1.04A} = 1346.2\ \Omega$

$\theta = \cos^{-1}\left(\frac{P_{\phi,SC}}{V_{\phi,SC} I_{\phi,SC}}\right) = \cos^{-1}\left(\frac{305W}{(1400)(1.04)}\right) = 77.91°$

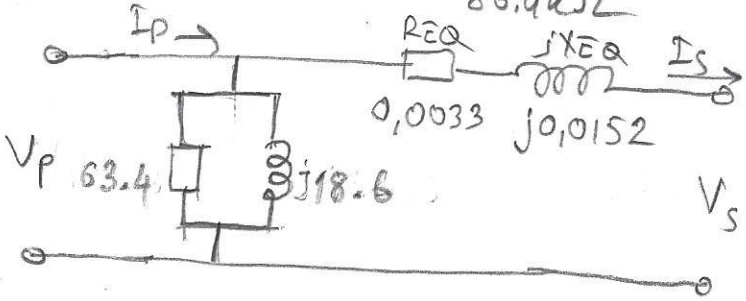
$Z_{EQ,P} = R_{EQ,P} + jX_{EQ,P} = 1346.2\angle 77.91 = 281.96 + j1316.34\ \Omega$

HVS

$Z_{base,P} = \frac{(V_{\phi base})^2}{S_\phi/3} = \frac{(24000)^2}{20kVA/3} = 86,400\ \Omega = 86.4k\Omega$

$$R_{EQ,P} = R_{EQ} = \frac{281{,}96\,\Omega}{86.4\,k\Omega} = 0{,}0033\ pu$$

$$X_{EQ,P} = X_{EQ} = \frac{1316{,}34\,\Omega}{86.4\,k\Omega} = 0{,}0152\ pu$$



b) 0,90 PF lagging $\Rightarrow -\cos^{-1}(0{,}88) = -28{,}36°$

$$V_P = V_S + I_S Z_{EQ} = 1{,}0\angle 0° + (1\angle -28{,}36°)(0{,}0033 + j\,0{,}0152)$$

$$V_P = 1{,}0\angle 0° + (1\angle -28{,}36)(0{,}00363\angle 24{,}73) = 1{,}0036\angle -0{,}01312\ pu$$

$$VR = \frac{V_{P,pu} - V_{S,pu}}{V_{S,pu}} \times 100 = \frac{1{,}0036 - 1.0}{1.0} \times 100 = 0{,}36\,\%$$

c) $P_{out_{pu}} = V_S I_S \cos\theta = (1.0)^2(1.0)(0{,}88) = 0{,}88\ pu$

$$P_{cu_{pu}} = I_S^2 R_{EQ} = (1.0)^2(0{,}0033) = 0{,}0033\ pu$$

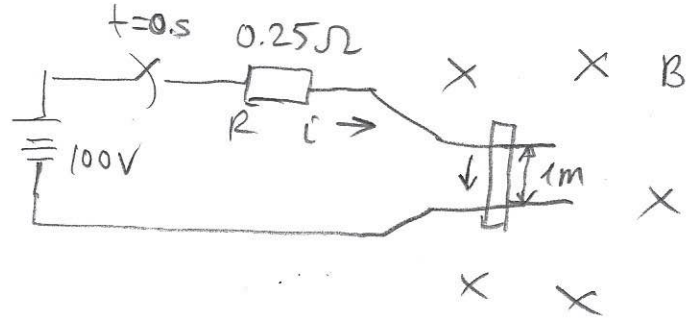
$$P_{core_{pu}} = \frac{V_P^2}{R_C} = \frac{(1.0036)^2}{53.4} = 0{,}01588$$

$$P_{in_{pu}} = P_{out} + P_{cu} + P_{core} = 0{,}88 + 0{,}0033 + 0{,}01588\ pu$$

$\Rightarrow P_{in} = 0{,}899$

$$\eta = \frac{P_{out}}{P_{in}} \times 100 = \frac{0{,}88}{0{,}899} \times 100 = 0{,}9786\,\%$$

2)



a) $F = ?$ $\quad i = \frac{V_B}{R} = \frac{100V}{0,25\Omega} = 400$, At starting

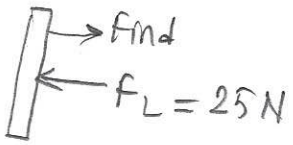
$F = i(\ell \times B) = i\ell B = (400)(1m)(0,5T) = 200N$, to the right (LHR)

b) At steady-state $\varepsilon_{ind} = V_B = v_{ss} B\ell \Rightarrow V_{ss} = \frac{V_B}{B\ell} = \frac{100V}{(0,5T)(1m)}$

$v_{ss} = ?$

$\Rightarrow V_{ss} = 200\ m/s$

c) $F_L = 25N$



$F_L = F_{ind} = i\ell B = 25N \Rightarrow i = \frac{25N}{(1m)(0,5T)} = 50A$

$\varepsilon_{ind} = V_B - iR = 100V - (50A)(0,25\Omega) = 87,5\ V$

$V_{ss,new} = \frac{\varepsilon_{ind}}{B\ell} = \frac{87.5V}{(0,5T)(1m)} = 175\ m/s.$

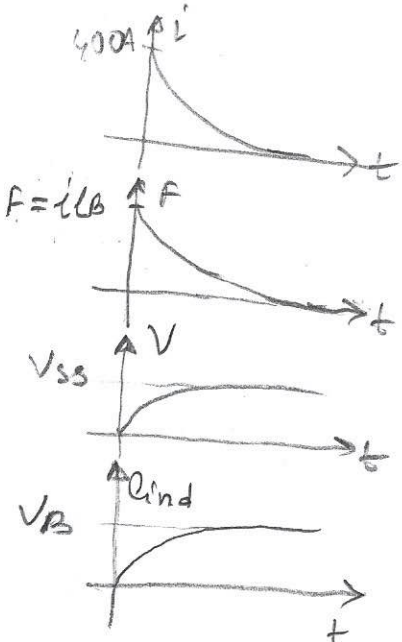
c) The efficiency

$P_{in} = V_B i = (100V)(50A)$

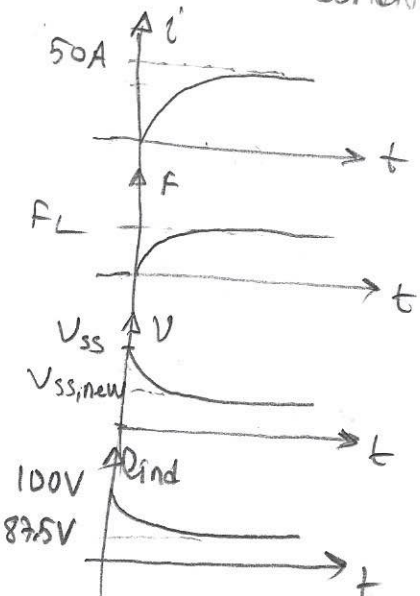
$P_{out} = F v = (25N)(175)$

$\eta = \frac{P_{out}}{P_{in}}(100) = 87.5\%$

d) At No load (b)



Under load condition (c)



3) 100-hp, 250-V, 1200 r/mm SHUNT DC MOTOR

$R_A = 0{,}03\,\Omega$, $R_F = 41.67\,\Omega$

No armature reaction, mechanical and Core losses negligible

$I_L = 126\,A$, $n_{initial} = 1103$ r/min

$I_A \rightarrow$ constant

n ? if R_F is raised to $50\,\Omega = R_{F2}$

① $I_{A1} = I_{L1} - I_{F1} = I_{L1} - \frac{V_T}{R_{F1}} = 126V - \underbrace{\frac{250V}{41.67\Omega}}_{6A} = 120\,A$

$I_{F1} = 6A \rightarrow E_{A1,MC} \cong 268V$ (from the magnetization curve)

② $I_{A1} \Rightarrow E_{A1,real} = V_T - I_{A1}R_A = 250V - (120A)(0{,}03\Omega) = 246.4V$ (real value under present condition)

③ $I_{F2} = \frac{V_T}{R_{F2}} = \frac{250V}{50\Omega} = 5A \rightarrow E_{A2} = 250V$ (from the curve)

④ Since I_A is constant, E_A remains constant, looking at ②

Hence $\frac{E_{A2}}{E_{A1}} = \frac{K\phi_2 n_{m2}}{K'\phi_1 n_{m1}} = 1 \Rightarrow n_{m2} = \frac{\phi_1}{\phi_2} n_{m1}$

(E_{A1} present condition)

⑤ $E_{A1} = E_{A2}$

$E_{A2,MC} = 250V$ at $I_{F2} = 5A$ and $E_{A1,MC} = 268$ at $I_{F1} = 6A$ (From the Fig 1) are directly proportional to the flux, now the ratio of $\frac{\phi_1}{\phi_2}$ will approximately equal to the ratio of $\frac{E_{A1,MC}}{E_{A2,MC}}$



$$\frac{\phi_1}{\phi_2} = \frac{E_{A1,MC}}{E_{A2,MC}} = \frac{268V}{250V} = 1.072$$, and from ④

$n_{m2} = (1.072)(1103\,r/min) = 1182.4\,r/min$

4) Sepately excited DC G. ∋ 172kW, 430V, 400A, 1800 r/min.

$R_A = 0{,}05\,\Omega$, $R_F = 20\,\Omega$, $R_{adj} = 0$ to $300\,\Omega$, $V_F = 430V$, $N_F = 1000$ turns/pole.

a) $R_{adj} = 63\,\Omega$ $n_{prime} = 1600$ r/min $V_{T,no\,load} = ?$

b) V_T if $I_L = 360A$, with compensating windings

c) V_T if $I_L = 360A$, without " "

$F_{AR} = 450$ A.turns

d) $I_{F,new} = ?$ to restore V_T to the original value (no load voltage) with existing compensating winding

$R_{adj} = ?$ (the required value of R_{adj})

a) $V_{T,no\,load} = E_{A,1600} \Rightarrow$ $I_f = \frac{V_F}{R_F + R_{adj}} = \frac{430V}{20\Omega + 63\Omega} = 5.18A$

$\frac{E_{A,1600}}{E_{A,1800}} = \frac{n_m}{n_o} \Rightarrow E_{A,1600} = \frac{n_m}{n_o} E_{A,1800} = \frac{1600}{1800}(430V)$

$= 382V$ (V_T no load)

$I_F = 5.18 \Rightarrow E_{A,1800} = 430V$ (from the MC (Fig 2)

$V_{T,NL} = 382V$

b) $V_T = E_A - I_A R_A = 382V - (360A)(0{,}05\Omega) = 364V$

c) $I_F^* = I_F - \frac{F_{AR}}{N_F} = 5.18 - \frac{450\text{ A.turns}}{1000\text{ turns/pole}} = 4{,}73A$

from the MC $E_{A0} = 410V$ at 1800 r/min

$E_{A,1600} = \frac{n_m}{n_o} E_{A0} = \frac{1600\text{ r/min}}{1800\text{ r/min}}(410V) = 364.4V$

$V_T = E_{A,1600} - I_A R_A = 364.4V - (360A)(0{,}05\Omega) = 346.4V$

4-) d) $I_F = ?$ to raise V_T to the no load value

$$E_{A,1600} = V_T + I_A R_A = 382V + (360A)(0,05\Omega) = 400V$$

$$\frac{E_{A,1600}}{E_{A,1800}} = \frac{n_m}{n_o} \Rightarrow E_{A,1800} = \frac{n_o}{n_m} E_{A,1600} = \frac{1800}{1600}(400V)$$

$$E_{A,1800} = 450V.$$

From the MC the corresponding $I_F = 6.15A$

$$I_F = 6.15 = \frac{V_{T,NL}}{R_F + R_{adj}} = \frac{430V}{20 + R_{adj}} \Rightarrow R_{adj} = \frac{430 - (6.15)(20)}{6.15}$$

$$= 49.9 \approx 50\Omega.$$